higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

DIGITAL ELECTRONICS N4

29 November 2019

This marking guideline consists of 8 pages.

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QUESTION 1

1.1	1.1.1	A		
	1.1.2	C		
	1.1.3	D		
	1.1.4	D		
	1.1.5	A		
			(5 × 1)	(5)
1.2	1.2.1	C		
	1.2.2	D		
	1.2.3	F		
	1.2.4	B		
	1.2.5	A		
			(5 × 1)	(5)
1.3	1.3.1	True		
	1.3.2	True		
	1.3.3	False		
	1.3.4	True		
	1.3.5	True		
			(5 × 1)	(5)

1.4 1.4.1

$$\begin{array}{r} 100111 = 011000 \\ \underline{\quad +1} \\ 011001 \end{array}$$

11001

1.4.2 10011 = 11010Gray

1.4.3 $D_{16} = 1101_2$

$B_{16} = 1010_2$

$$\begin{array}{r} 1101 \\ \underline{+1010} \\ 10111 \end{array}$$

(3 × 2) (6)

[21]

QUESTION 2

2.1 D, 7_{16} = 1101, 0111

3, 6_8 = 0011, 1100 = 1100, 0011

```
  1101, 0111
  1100, 0011
-------------
  1001, 1010
        +1
-------------
  1001, 1011
```

=9, 6875_{10}

2.2 7C, B_{16} = 01111100, 1011

$6,5_8$ = 110, 101

```
                 10010,110
1101010√11111001011000
      -1101010
         10010101
         -1101010
            10101110
            -1101010
             10001000
             -1101010
                 111100
```

=22, 6_8

2.3 86,E_{16} = 10000110, 1110

7, 5_{10} = 111, 1

```
      100001101110
           1111
-------------------
      100001101110
     100001101110
    100001101110
100001101110
-------------------
1111110011, 10010
```

= 3F3, 9_{16}

(3 × 6) **[18]**

QUESTION 3

3.1 $F = A\,\overline{B}\,\overline{C} + A\,\overline{B}\,\overline{C} + ABC + \overline{A}\,B\,\overline{C}$

$A\,(\overline{B}\,C + B\,\overline{C}) + \overline{A}\,(B\overline{C} + \overline{B}\,C)$

$A\,\overline{(B \quad C)} + \overline{A}\,(B \quad C)$

LET $B \quad C = K$

$A\,\overline{K} + \overline{A}\,K$

$A \quad K$

$A \quad B \quad C$

3.2 $\overline{\qquad\qquad}$

$F = \overline{A}\,\overline{B} + \overline{A}B + A\,\overline{B}$

$\overline{(\overline{A}\,B)}\ \overline{(\overline{A}\,B)}\ \overline{(A\,\overline{B})}$

$(A + \overline{B})\,(A + \overline{B})\,(\overline{A} + B)$

$AA + A\overline{B} + A\overline{B} + \overline{B}\,\overline{B}\,(\overline{A} + B)$

$A\,(1 + \overline{B} + \overline{B}\,) + \overline{B}\,(\overline{A} + B)$

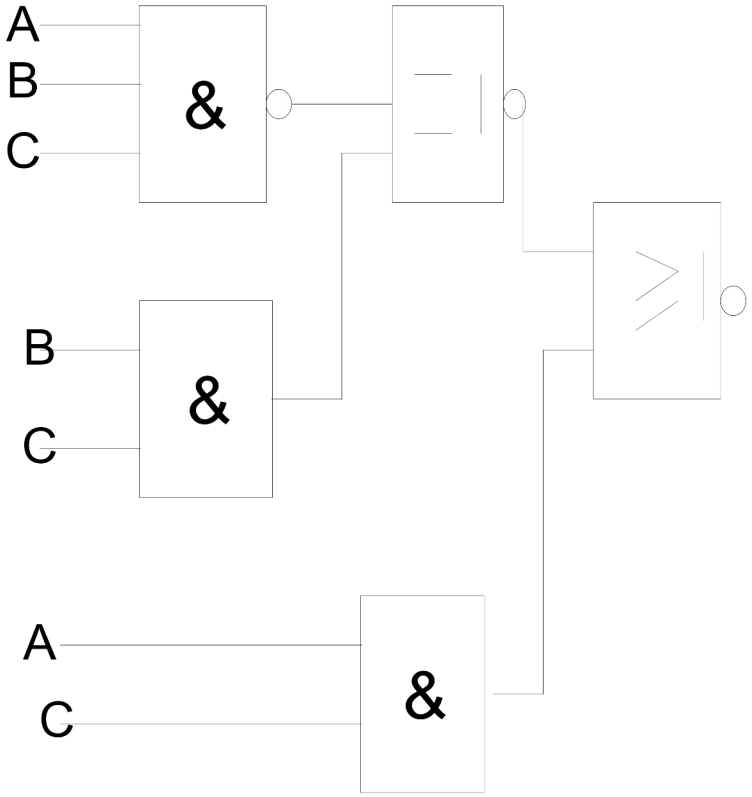
$A + \overline{B}\,(\overline{A} + B)$

$A\,\overline{A} + A\,B + \overline{A}\,\overline{B} + B\,\overline{B}$

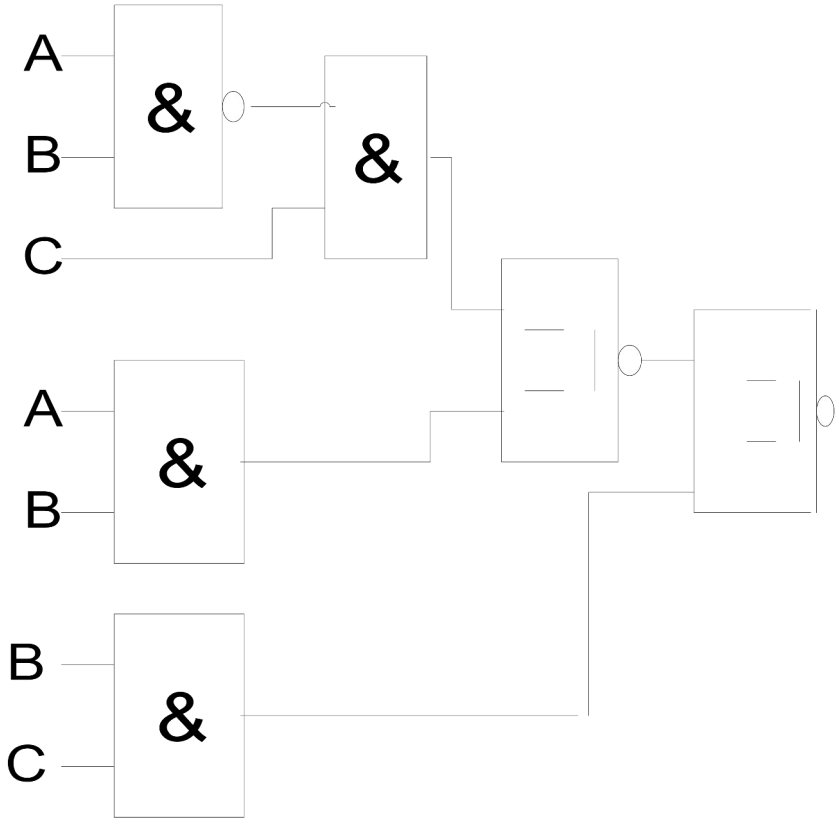
AB

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3.3



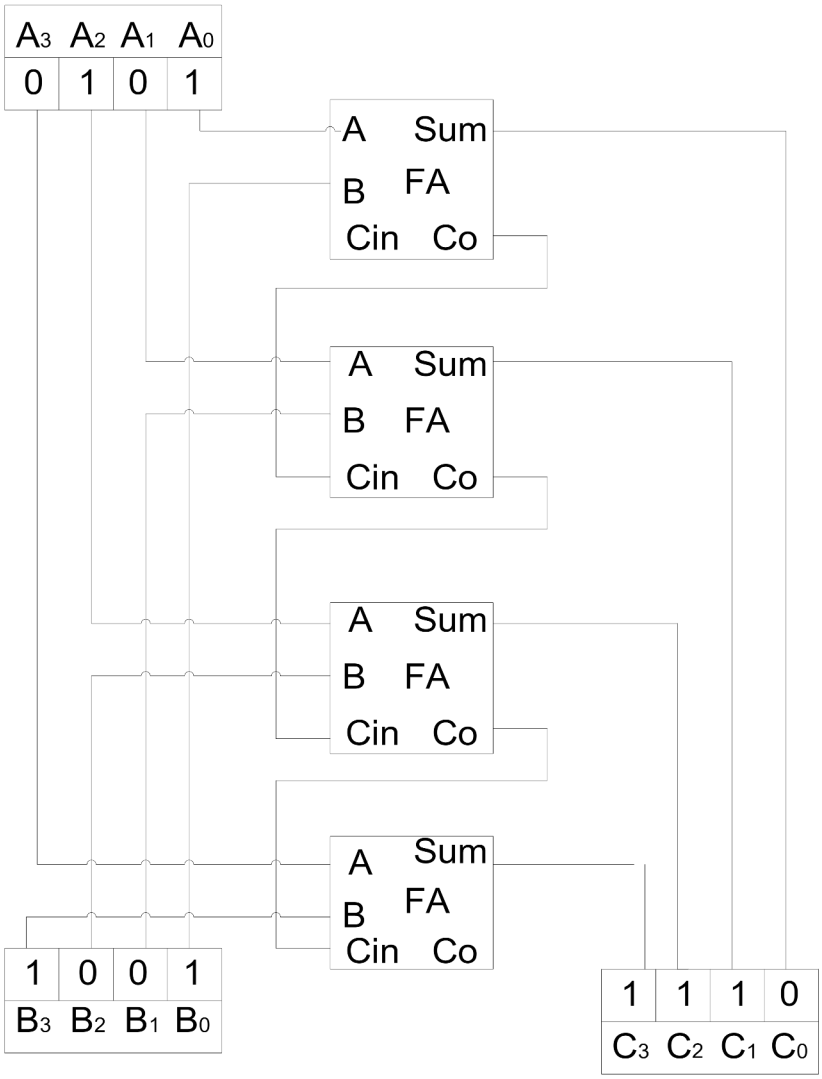
3.4



(4 × 6) **[24]**

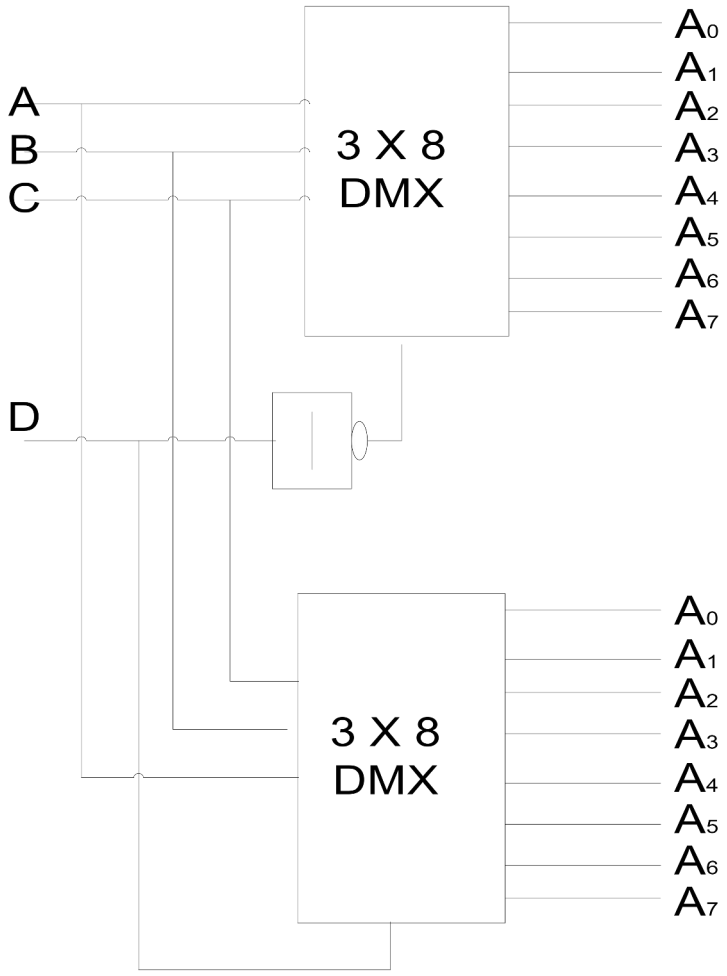
QUESTION 4

4.1



(10)

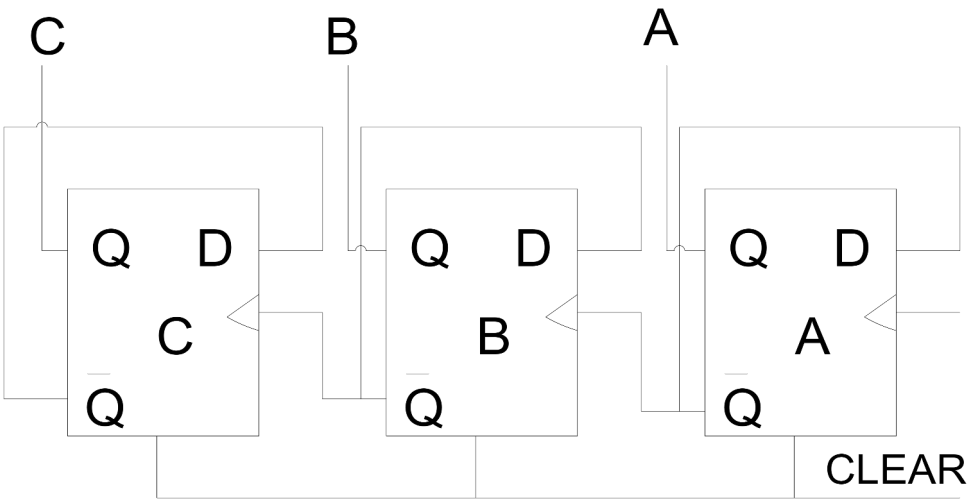
4.2



(9)
[19]

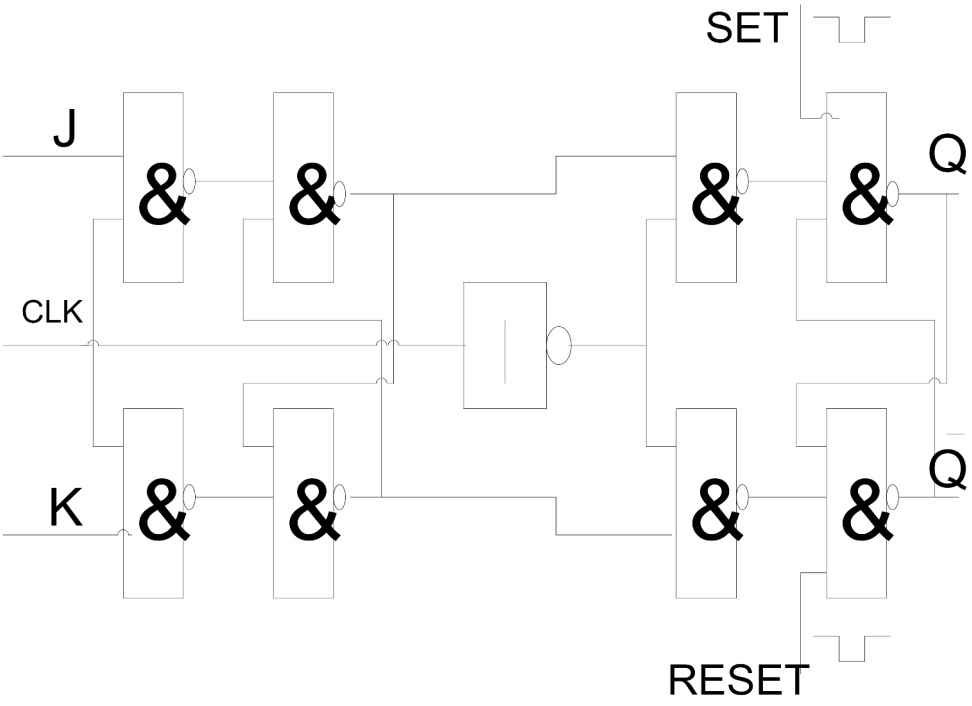
QUESTION 5

5.1



(8)

5.2



(10)
[18]

TOTAL: 100